iBookTrack — System Functionalities

> *Prepared for management review. This document lists all functionalities currently implemented in the iBookTrack system.*

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1. What the System Does

iBookTrack is a Warehouse Management System (WMS) and Executive Dashboard built for Innodata's physical book digitization workflow. It replaces manual Excel-based tracking with real-time, database-backed tracking of every physical book as it moves through digitization.

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2. The 8-Stage Book Pipeline

Every book/box is tracked sequentially through these stages, with a digital handshake required to move from one stage to the next (so nothing can be lost or skipped):

#	Stage	Function
1	**Warehouse**	Books are received and scanned into the system; immediately checked for processability (damage, missing pages, etc.)
2	**Inventory**	Confirms official acceptance of the books into inventory
3	**Cut**	Books are sent to the cutting station to remove bindings
4	**Endorsement (QA)**	Full quality inspection — checked for missing pages, blurriness, physical damage
5	**Scan**	Books are handed off to the high-speed scanning machines
6	**Transmit**	Confirms the digital PDFs have successfully reached the final servers
7	**Storage**	Books are placed into their final physical storage bins

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3. Core Functionalities

Authentication & User Management

- Secure login/logout with JWT-based sessions and automatic token refresh
- User registration and profile management
- Password change
- Role-based access control (e.g., Supervisor vs. operational staff roles) — restricts who can view data or perform actions
- Single sign-on integration with Innodata's internal system (iLMS)
- Rate-limiting on login/registration to prevent abuse

Warehouse Receiving

- Start, resume, and complete scanning sessions per lot
- Barcode/QR scanning for fast book intake
- Automatic detection of duplicate/similar lots and lot-number mismatches against the iLMS catalog
- Bulk-save of scanned sessions

- Ability to flag books as damaged/non-processable at the point of intake, with a reason code
- Lock mechanism so only one user can work a box/session at a time (prevents double-entry); other users see it live as "locked"
- Live warehouse overview showing all active lots and their scan status

Inventory & Box Tracking

- View and search all boxes by number, lot, or ID
- Advance a box to the next pipeline stage or send it forward
- Roll back a box to a previous stage if an error is discovered
- Receive boxes at each checkpoint (Inventory, Cut, Scan handshakes)
- Track pipeline transition history (audit trail of every stage change)
- Cross-check books against Simflex machine data for scanning verification
- Print box labels

Lot Management

- View all lots and drill into details for a specific lot
- Transmission summary per lot (how many books transmitted vs. pending)

Quality Control (QC) / Flagging

- Flag or unflag individual books during intake or QA with a specific issue reason (torn pages, bleed-through, missing pages, blurriness, etc.)
- Central "Missing Item Log" of all flagged/rejected books with reasons
- Resolve QC flags once addressed
- Export mismatched/flagged items for reporting
- Detect and warn about missing ISBNs in the catalog

Scanning Floor Monitoring

- Live view of scanning activity by machine
- Per-machine book assignment and throughput ranking
- Export scan data
- Cross-reference against Simflex (the physical scanning machine system) for status validation
- Enrich scan records with iLMS metadata (titles, etc.)

Storage / Warehouse Locator

- Provision and manage physical storage bin locations (Supervisor only)
- Assign/confirm/remove batches of books to storage locations
- Instantly search and locate any book or lot's physical storage bin
- View boxes ready for storage assignment and pending assignments

Executive Dashboard

- Live stats: active lots, books currently in the pipeline, daily transmissions, flagged/rejected items
- Visual pipeline breakdown (how many books are at each of the 8 stages)
- Recent activity feed of live scans
- Global search (look up any book/lot instantly)
- Real-time notifications (via live event stream) when a station is locked, unlocked, or delayed — so staff know instantly if a session is stuck

Reporting

- Real-time lot status reports
- Daily output reports

- Mismatch/QC export reports for offline review

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4. Technology Summary (for IT reference)

Layer	Technology
Frontend	Next.js 15, React 19, Tailwind CSS
Backend	Node.js, Express.js (REST API)
Database	PostgreSQL
Authentication	Custom JWT-based security
Real-time updates	Server-Sent Events (SSE)
External Integrations	Innodata iLMS (catalog/SSO), Simflex (scanning machines)

For full technical documentation (API routes, database schema, deployment steps), see the companion documents in `docs/`: `PROJECT_OVERVIEW.md`, `architecture/BACKEND.md`, `architecture/FRONTEND.md`, `guides/MIGRATION_GUIDE.md`.